ANY CONCEPCION GONZALEZ RODRIGUES

Molecular Bioengineer | Neural Electrophysiology, HD-MEA & Cell Culture
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RESEARCH PROFILE

Molecular Bioengineering graduate with research experience in retinal electrophysiology, neural tissue preparation and neuronal cultures. During my Master's thesis, I established and optimised an *ex vivo* HD-MEA workflow to investigate spontaneous and visually evoked activity in adult zebrafish retina. My research interests centre on retinal circuit function, disease-related changes in neuronal activity, and the functional characterisation of human retinal models, including organoids.

EDUCATION

MSc in Molecular Bioengineering | University of Porto - ICBAS/FEUP
Sep 2022 - Oct 2024 | Porto, Portugal
Thesis: Establishing CMOS-MEA Recordings in Adult Zebrafish Retina
Supervisor: Prof. Dr. Michael Brand - CRTD, Germany

BSc in Bioengineering | University of Porto - ICBAS/FEUP
Sep 2018 - Jun 2022 | Porto, Portugal
Thesis: Development of Optical Sensors Using Molecularly Imprinted Polymers for Ferritin Detection
Supervisor: Dr. Margarida Piloto - ISEP, Portugal

RESEARCH EXPERIENCE

Master's Thesis Researcher – Neural Electrophysiology
Center for Regenerative Therapies Dresden (CRTD), TU Dresden | Feb 2024 – Aug 2024

- Established and optimised an ex vivo HD-MEA recording workflow for adult zebrafish retina, including retinal dissection, tissue positioning, perfusion and electrode–tissue contact.
- Investigated spontaneous and visually evoked retinal activity using constant and chirp light stimulation protocols and pharmacological modulation with meclofenamic acid (MFA).
- Performed spike sorting using 3BrainWave and analysed neuronal firing rates, PSTHs and stimulus-evoked responses using R and Python.
- Troubleshot tissue preparation and recording conditions, documenting experimental findings through scientific figures, presentations and a Master's thesis.

Research Intern – Neural Cell Culture and Co-culture
German Center for Neurodegenerative Diseases (DZNE), Dresden | Sep 2023 – Feb 2024

- Maintained human and rat olfactory cell cultures and olfactory bulb–epithelium co-cultures under sterile conditions.
- Performed MEA recordings to investigate neuronal activity and network responses following odorant stimulation (eugenol and isoamyl acetate) and pharmacological modulation (IBMX and forskolin).
- Prepared culture media and experimental solutions and contributed to immunofluorescence and confocal microscopy workflows.
- Analysed electrophysiological parameters, including firing rates, burst activity and inter-burst intervals, and documented experimental results.

Research Intern
BioMark, ISEP, Porto | Feb 2022 – Jul 2022

- Contributed to optical sensor development for ferritin detection using molecularly imprinted polymers and UV–Vis spectroscopy

LABORATORY & TECHNICAL SKILLS

Electrophysiology: HD-MEA recordings, extracellular neuronal activity, retinal tissue preparation, visual and pharmacological stimulation, experimental troubleshooting.

Cell Culture & Microscopy: Aseptic cell culture, olfactory epithelium and olfactory bulb cultures, co-cultures, immunofluorescence, confocal microscopy.

Data Analysis: Python, R, spike-train analysis, PSTH, firing-rate analysis, statistical comparisons, data visualisation.

Instrumentation & Software: 3Brain BioCAM Duplex, Accura 4096, BrainWave, Axion Maestro.

Molecular Biology: DNA extraction, conventional PCR, agarose gel electrophoresis, gel imaging.

PROFESSIONAL EXPERIENCE

Account Management Intern
MaxWell Biosystems, Zurich | May 2025 – Apr 2026

- Supported international scientific customers working with HD-MEA platforms for neuronal cultures, organoids and neural tissue.
- Communicated electrophysiology applications through scientific webinars, conferences and digital content.
- Developed an Ambassadors and Advocates initiative, analysing over 250 accounts to identify research collaborators and engagement opportunities.

TRAINING AND CERTIFICATIONS

- FELASA-accredited laboratory animal training LAS modules C01 and F09-F11
- Zebrafish handling course
- Introduction to Python and Machine Learning for the Biosciences (i3S)

LANGUAGES

Portuguese — native (C2); English — C1; Spanish — B1; German — A1, actively learning

ACADEMIC REFERENCES

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